

ATHARVA PATIL

Software Engineer · Founder · Product Builder

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Portfolio — atharvapatil.tech

PROFESSIONAL SUMMARY

Software engineer and product builder focused on **desktop applications, developer tools, operating systems, and creative technology**. Founder of **Northbyte Studios**, an independent software studio building privacy-first, practical products with zero telemetry and long-term maintainability. Ships end-to-end — product direction, interface design, engineering, testing, documentation, release. Currently building an encrypted vault, two Linux operating systems, and a general-purpose programming language, all under one studio and one pair of hands.

PROJECTS

Nexus · Encrypted Personal Vault

Founder, Lead Engineer, Designer — github.com/p4inz-code/nexus-desktop · Jun 2026 – Present · Windows Desktop

- **Cryptographic core:** AES-256-GCM authenticated encryption over an NXG1 container format, Argon2id key derivation, Windows Hello secondary unlock, in-app SHA-256-verified updater, portable-mode operation.
- **Trust posture:** zero telemetry, zero cloud, zero sync. Every byte encrypted at rest, decrypted only in memory while the vault is unlocked.
- **v11 UI overhaul:** entire visual system rebuilt across lock screen, media browser, credential + note detail, settings, updater. Cryptographic architecture unchanged — v11 is a design ship, not a security ship.
- **Stack:** C# / .NET 8 / WPF · AES-256-GCM · Argon2id · GitHub Actions CI.

Mission OS · Privacy-First Linux Distribution

Founder, Designer, Systems Engineer — github.com/p4inz-code/mission-os · 2026 – Present · Linux, GPLv3

- **Four pillars applied to the OS itself:** privacy-first, security-conscious, portable-first, offline-first. Built on Debian Stable + KDE Plasma; installer authored in QML / Qt.
- **Installer discipline:** every advanced option (LUKS2, Secure Boot, TPM-sealed unlock, recovery keys) explains its benefits, limitations, compatibility, and trade-offs in the same view as the toggle.
- **141 / 141 pixel checks** passed on installer foundation. 14 CTest tests, 13 runtime suites. Accessibility, keyboard, theme, and reduced-motion all validated pre-ISO.
- **Interactive concept demo** live at atharvapatil.tech/mission-os/demo — boot sequence + desktop with six real informational apps.

Kanvaz · Visual Reference Workspace

Founder, Sole Engineer + Designer — github.com/p4inz-code/kanvaz · Jun 2026 – Aug 2026 · Electron, MIT

- **Plugin system** (window.KanvazPluginAPI) with main-process consent — plugins can register card types, ship alternate themes, add Settings panels. Theme Creator ships as the reference plugin.
- **Container format:** safer .kanvaz zip archive with per-asset SHA-256 integrity hashes. Detects tampering on load.
- **Final release: v4.2.2.** Development wound down cleanly; no v5 planned. Free forever, open source.

Pursue OS · Investigation-Focused Linux OS

Founder, Sole Developer — github.com/p4inz-code/pursue-os · 2026 – Present · Linux, Apache 2.0

- **Purpose-built for** OSINT, DFIR, secure research, evidence collection, and investigation workflows.
- **Two flagship surfaces:** Investigation Terminal + Investigation Browser with integrated Tor. AI is assistant-not-authority — never performs detection, only explains.

Veris · Security Investigation Platform

Founder, Sole Engineer — github.com/p4inz-code/veris · Jun 2026 – Present · Windows, Pre-v1.0.0

- **Deterministic analysis engine — no black-box AI scoring.** AI is intentionally limited to a read-only explanation layer over the deterministic core.
- **3,100+ automated tests.** Offline-first — no analysis leaves the machine.
- **Stack:** TypeScript / Node.js / SQLite, modular monorepo.

Obscura, MINK, repo-map · Additional shipped / in-flight work

Sole engineer across all three

- **Obscura** (Lua AST toolkit for Roblox) — v1.0.0 shipped with 340 / 340 tests passing. MIT, free forever.
- **MINK** — general-purpose programming language, in design. Simplicity, safety, reliability.
- **repo-map** — Engineering Intelligence CLI. Published on npm as @p4inz-code/repo-map. 870+ tests. On hold at v2.2.0.

EXPERIENCE

Northbyte Studios — Founder & Independent Software Studio

2026 – Present · Navi Mumbai, India (remote-friendly)

- Independent software studio developing desktop applications, developer tools, and open-source systems.
- **Full-cycle ownership:** product direction, architecture, engineering, UI / UX, testing + QA, documentation, branding, release engineering — all performed by one person.
- Ships every product against a fixed principle stack: **no telemetry, no cloud lock-in, explain the trade-off on-screen, free-forever when it can be, one person whole product.**

Freelance — Web + UX + Brand

2025 – Present

- Independent client work: web development, UI / UX, visual identity, 3D / VFX visualization. Notable engagement includes a full production website build for a performing-arts academy.
- **Scope shapes:** short-loop (landing / marketing sites — 1–2 weeks, fixed price); product build (dashboards, product surfaces — 4–8 weeks, discovery week first); studio work (3D renders, lighting studies, promo animation — shot-scoped).

TECHNICAL SKILLS

Languages	C#, C / C++, TypeScript, JavaScript, HTML, CSS, Lua / Luau, Bash
Frameworks & Platforms	.NET 8, WPF, Avalonia UI, Electron, Node.js, Astro, Qt / QML, Debian Linux, Windows
Cryptography & Security	AES-256-GCM, Argon2id, LUKS2, TPM, Secure Boot, Windows Hello, deterministic analysis engines
3D / VFX	Blender, Substance, DaVinci Resolve, lighting studies, product renders, motion / camera work
Tools & Workflow	Git / GitHub / Actions, VS Code, Visual Studio, FFmpeg, npm / pnpm, Cloudflare Pages
Areas	Desktop application development, product development, UI / UX, developer tooling, open source, Linux, privacy & security, creative technology, AI-assisted software

EDUCATION

B.Sc. Animation

D. Y. Patil University, Nerul, Navi Mumbai · 2025 – 2028 · Second year in progress (2026 – 2027)

PHILOSOPHY

Build software that is useful, understandable, reliable, and worth maintaining. Deliberate product decisions. Clean interfaces. Privacy by default. Practical engineering. Long-term maintainability.

Five tenets that show up in every product I ship: No telemetry, ever. Explain the trade-off on-screen. Content-accurate over content-marketed. One person, whole product. Free-forever when it can be.

Résumé maintained as canonical Markdown at atharvapatil.tech/resume/Atharva_Patil_Resume.md · Every fact verifiable against the corresponding GitHub repository or the live status page at atharvapatil.tech/status.